

THE TWENTY-FOURTH WILLIAM LOWELL PUTNAM MATHEMATICAL COMPETITION

December 7, 1963

Morning Session

1. (i) Show that a regular hexagon, six squares, and six equilateral triangles can be assembled without overlapping to form a regular dodecagon.

(ii) Let $P_1, P_2, \dots, P_{12}$ be the successive vertices of a regular dodecagon. Explain how the three diagonals $P_1 P_9, P_2 P_{11}$, and $P_4 P_{12}$ intersect.

(page 569)

2. Let $\{f(n)\}$ be a strictly increasing sequence of positive integers such that $f(2) = 2$ and $f(mn) = f(m)f(n)$ for every relatively prime pair of positive integers m and n (the greatest common divisor of m and n is equal to 1). Prove that $f(n) = n$ for every positive integer n .

(page 570)

3. Find an integral formula for the solution of the differential equation

$$\delta(\delta - 1)(\delta - 2) \cdots (\delta - n + 1)y = f(x), \quad x \geq 1,$$

for y as a function of x satisfying the initial conditions $y(1) = y'(1) = \cdots = y^{(n-1)}(1) = 0$, where f is continuous and

$$\delta \equiv x \frac{d}{dx}. \quad \text{(page 571)}$$

4. Let $\{a_n\}$ be a sequence of positive real numbers. Show that

$$\limsup_{n \rightarrow \infty} n \left(\frac{1 + a_{n+1}}{a_n} - 1 \right) \geq 1.$$

Show that the number 1 on the right-hand side of this inequality cannot be replaced by any larger number. (The symbol $\limsup$ is sometimes written $\underline{\lim}$.)

(page 572)

5. (i) Prove that if a function f is continuous on the closed interval $[0, \pi]$ and if

$$\int_0^\pi f(\theta) \cos \theta \, d\theta = \int_0^\pi f(\theta) \sin \theta \, d\theta = 0$$

then there exist points α and β such that

$$0 < \alpha < \beta < \pi \quad \text{and} \quad f(\alpha) = f(\beta) = 0. \quad (\text{page 574})$$

(ii) Let R be any bounded convex open region in the Euclidean plane (that is, R is a connected open set contained in some circular disk, and the line segment joining any two points of R lies entirely in R). Prove with the help of part (i) that the centroid (center of gravity) of R bisects at least three distinct chords of the boundary of R . (page 574)

6. Let U and V be any two distinct points on an ellipse, let M be the midpoint of the chord UV , and let AB and CD be any two other chords through M . If the line UV meets the line AC in the point P and the line BD in the point Q , prove that M is the midpoint of the segment PQ . (page 575)

Afternoon Session

1. For what integer a does $x^2 - x + a$ divide $x^{13} + x + 90$? (page 577)

2. Let S be the set of all numbers of the form $2^m 3^n$, where m and n are integers, and let P be the set of all positive real numbers. Is S dense in P ? (page 578)

3. Find every twice-differentiable real-valued function f with domain the set of all real numbers and satisfying the functional equation

$$(f(x))^2 - (f(y))^2 = f(x + y)f(x - y)$$

for all real numbers x and y . (page 579)

4. Let C be a closed plane curve that has a continuously turning tangent and bounds a convex region. If T is a triangle inscribed in C with maximum perimeter, show that the normal to C at each vertex of T bisects the angle of T at that vertex. If a triangle T has the property just described, does it necessarily have maximum perimeter? What is the situation if C is a circle? (A convex region is a connected open set such that the line segment joining any two points of the set lies entirely in the set.) (page 580)

5. Let $\{a_n\}$ be a sequence of real numbers satisfying the inequalities

$$0 \leq a_k \leq 100a_n \quad \text{for } n \leq k \leq 2n \quad \text{and} \quad n = 1, 2, \dots,$$

and such that the series

$$\sum_{n=0}^{\infty} a_n$$

converges. Prove that

$$\lim_{n \rightarrow \infty} na_n = 0. \quad (\text{page 582})$$

6. Let E be a Euclidean space of at most three dimensions. If A is a nonempty subset of E , define $S(A)$ to be the set of all points that lie on closed segments joining pairs of points of A . For a given nonempty set A_0 , define $A_n \equiv S(A_{n-1})$ for $n = 1, 2, \dots$. Prove that $A_2 = A_3 = \dots$. (A one-point set should be considered to be a special case of a closed segment.)

(page 583)